MANOLO Innovation Lab

Pre-workshop briefing



Smaller, faster, explainable

Model compression and explainable AI for trustworthy deployment



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Two key terms

What explainable AI and model compression mean

Explainable AI (XAI)

Methods that show why a model made a prediction, in a form a person can inspect.

What in the image led to this decision?



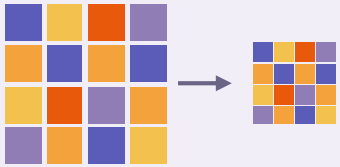
Warm colours mark the regions that most influenced the prediction.

In the notebook: Grad-CAM colours the regions that most raised the score of the predicted class: a heatmap.

Model compression (MC)

Techniques that make a trained model smaller and faster, while keeping its accuracy close to the original.

Can it run on a phone, a drone or a small sensor?



Original

Compressed

Same task, with less memory, time and energy.

In the notebook: Quantization stores each number with fewer bits. **Pruning** removes the weights that contribute least.

Next: why both matter, for legal, ethical and operational reasons.

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Legal: what the EU AI Act asks

Regulation (EU) 2024/1689 is risk-based: obligations scale with the intended purpose

Risk categories

Unacceptable risk
Prohibited · Art. 5

High risk
Strict requirements · Art. 6

Transparency risk
Disclosure duties · Art. 50

Minimal risk
No new obligations

Requirements for high-risk systems

Transparency Art. 13
Deployers must be able "to interpret a system's output and use it appropriately".

Human oversight Art. 14
Overseers must "correctly interpret" the output, with the interpretation tools available.

Accuracy, robustness Art. 15
Consistent "throughout their lifecycle": re-validate after compressing.

Documentation Art. 11, Annex IV
Records the computational resources used and how outputs can be interpreted.

Right to explanation Art. 86
Affected people can obtain "clear and meaningful explanations".

Penalties Art. 99
Up to €15 M or 3 % of worldwide annual turnover.

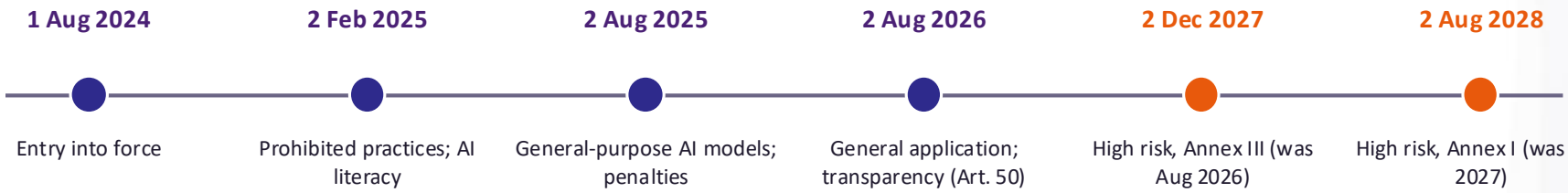
The Act requires transparency and human oversight, not a specific XAI method [4]. XAI is a practical means to meet these requirements.

Energy use is also addressed: Art. 40(2) standards and Annex XI documentation. Sources: [5] Regulation (EU) 2024/1689 · [4] Panigutti et al., FAccT 2023

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Legal: when it applies, and to what

Obligations phase in. The Digital Omnibus on AI, Regulation (EU) 2026/1744, moved the high-risk dates



Would your model be high-risk?

Risk follows the intended purpose, not the architecture

	**Chest X-ray**	Diagnostic support software	High risk: medical device (Annex I), from 2 Aug 2028
	**Wildfire**	Monitoring aid or dispatch priority	Minimal as a monitoring aid; high if it prioritises firefighter dispatch (Annex III, 5(d))
	**Rice blast**	Advice to farmers	Minimal risk

Illustrative, not legal advice. Sources: [5] Regulation (EU) 2024/1689, Art. 6, Art. 113, Annexes I and III · [6] Regulation (EU) 2026/1744 (Digital Omnibus on AI)

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Ethics: trustworthy beyond compliance

The EU Ethics Guidelines for Trustworthy AI set seven requirements [7]; three are at stake here

Transparency and accountability

"Clever Hans" models score well by exploiting artefacts, such as a watermark, instead of the object [8]. COVID-19 X-ray models relied on confounders; XAI should be "a prerequisite to clinical deployment" [9].

Fairness

Compression can keep top-line accuracy while hurting rare, atypical and under-represented classes disproportionately [10, 11, 12].

Environmental well-being

Training and running models costs energy and carbon [13]. Model, data centre and processor choices change it by up to 100 to 1000 times [14]. Green AI: report efficiency next to accuracy [15].

Human agency & oversight | Robustness & safety | Privacy & data governance | **Transparency** | **Diversity & fairness** | **Societal & environmental well-being** | **Accountability**

[7] HLEG 2019 · [8] Lapuschkin 2019 · [9] DeGrave 2021 · [10, 11] Hooker 2019, 2020 · [12] Paganini 2020 · [13] Strubell 2019 · [14] Patterson 2021 · [15] Schwartz 2020

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Operational: fit where it runs

Edge deployment brings lower latency, privacy and autonomy, at the cost of tight memory, compute and energy budgets

Why run on the device [16, 17]

Latency
decisions in real time

Bandwidth
send alerts, not images

Privacy
data stays where it is produced

Connectivity
fields and forests are often offline

In our domains

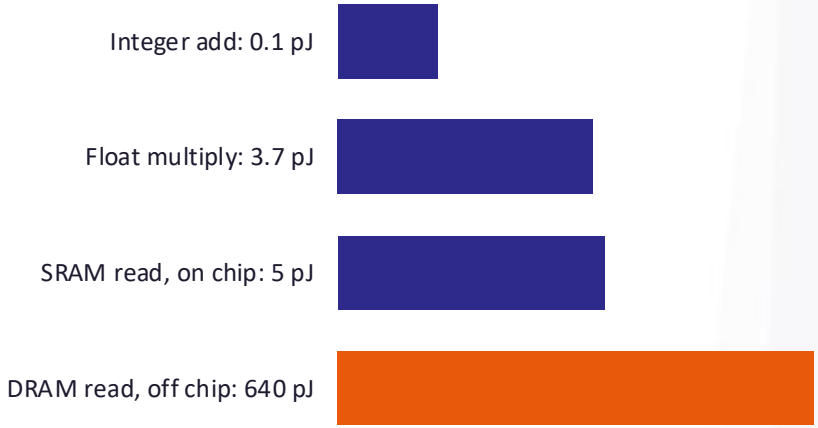


Wildfire: an edge CNN classifies each image in about 30 ms and sends only alerts [18].



Rice blast: a handheld device diagnoses crops in remote fields with limited connectivity, using a pruned model [19].

Energy per 32-bit operation, 45 nm CMOS (log scale) [20]



One DRAM read ≈ 6400 integer additions. Fewer, smaller weights mean less memory traffic.

[16] Chen & Ran, Proc. IEEE 2019 · [17] Shuvo et al., Proc. IEEE 2023 · [18] Almeida et al., IEEE TII 2022 · [19] Joshi et al., IEEE Sensors J. 2022 · [20] Horowitz, ISSCC 2014, as tabulated by Han et al. 2015

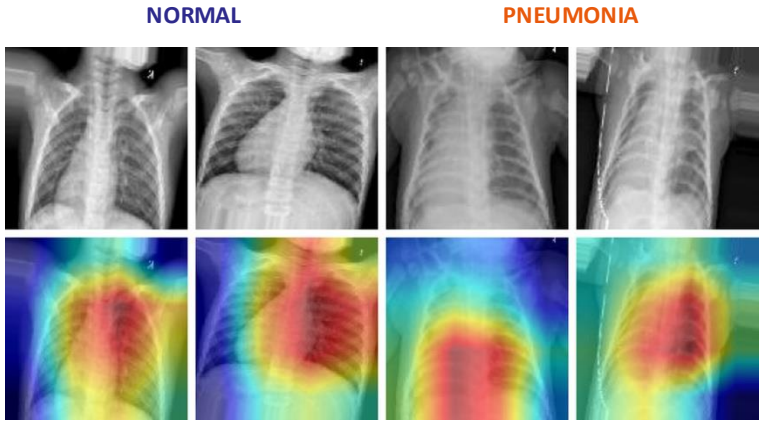
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XAI: Grad-CAM

Which regions of the image support class c? [21]

Input image → ResNet-18 forward pass → layer4 maps A^k $512 \times 8 \times 8$ → Back-propagate score y^c → Average each gradient: α_k^c → $\mathrm{ReLU}(\Sigma\, \alpha_k^c\, A^k)$ upsample, overlay

$$\alpha_k^c = (1/Z) \sum_i \sum_j \partial y^c / \partial A^k_{ij} \qquad L^c_{\text{Grad-CAM}} = \mathrm{ReLU}\left(\sum_k \alpha_k^c\, A^k \right)$$



Read the heatmap with care

Coarse. layer4 is 8×8 at 256 px and 4×4 at 128 px.

Where, not why. A heatmap is evidence to examine, not proof.

Test the explainer. Some saliency maps barely change when the weights are randomised [22]. In radiology, all eight methods tested failed at least one trust criterion [23].

Notebook example (AugCXR): input above, Grad-CAM below.

[21] Selvaraju et al., ICCV 2017 · [22] Adebayo et al., NeurIPS 2018 · [23] Arun et al., Radiology: AI 2021 · Library: pytorch-grad-cam

The compression toolbox

Four families of techniques [24]; the notebook uses the first two

Notebook 7.1

Quantization

Fewer bits per number: FP32 → INT8.

Notebook 7.2

Pruning

Remove weights or whole filters.

Knowledge distillation

A small student learns to imitate a large teacher.

Low-rank and lean design

Factorise layers, or design lean architectures (e.g. MobileNet).

49×

VGG-16: 552 MB → 11.3 MB

Pruning, quantization and Huffman coding combined, with no loss of accuracy. AlexNet: 35× (240 MB → 6.9 MB) [3].

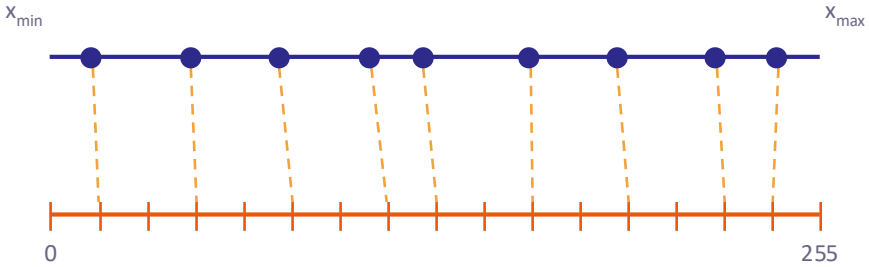
[24] Li et al., Computers 2023 · [3] Han et al., ICLR 2016

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Quantization: FP32 to INT8

Store and compute with 8-bit integers instead of 32-bit floats [25, 26]

FP32: continuous values



INT8: 256 integer levels (17 shown)

$$x \approx s\,(q - z) \qquad q = \text{clamp}(\,\text{round}(x / s) + z\,)$$

s: scale, *z*: zero point, both chosen from the range of the values. The gap between a dot and its tick is the rounding error.

Post-training static quantization (notebook 7.1)

Fuse → **Observe** → **Calibrate** → **Convert**

Fuse Conv + BN + ReLU, insert observers, run a few training batches to choose s and z, then convert to int8.

4×	≤ 2 %	2-3×
smaller weights	typical accuracy loss	faster on CPUs

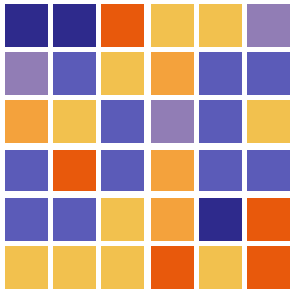
- Our ResNet-18: about 45 MB → about 11 MB.
- PyTorch runs these int8 kernels on the CPU only; the speed-up depends on the hardware.

[25] Jacob et al., CVPR 2018 · [26] Krishnamoorthi, arXiv 2018 (accuracy and speed-up figures)

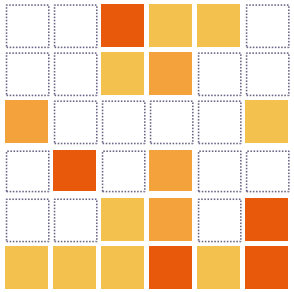
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Pruning: remove what contributes least

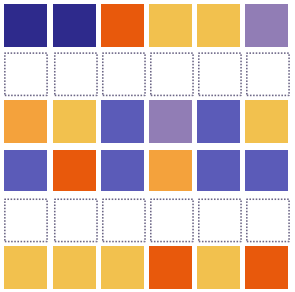
Magnitude pruning, then fine-tuning [27]



Dense



Unstructured: scattered zeros



Structured: whole filters removed

Magnitude criterion

Rank the weights by |w|, zero the smallest fraction (globally across layers in the notebook), then fine-tune. AlexNet: 9× fewer parameters, VGG-16: 13×, no loss of accuracy [27].

Zeros are not free

Unstructured zeros stay in dense tensors: the file size and the latency do not change without sparse formats or hardware [28].

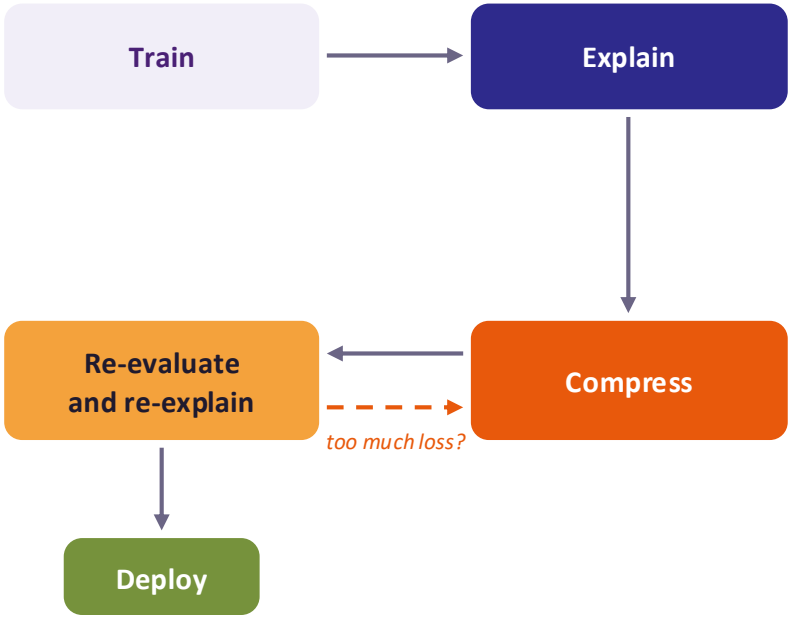
Structured is faster

Removing whole filters shrinks the tensors and gives real speed-ups [28, 29]. Compare fairly: report sparsity, size, latency and accuracy together [30].

[27] Han et al., NeurIPS 2015 · [28] Li et al., ICLR 2017 · [29] He & Xiao, IEEE TPAMI 2023 · [30] Blalock et al., MLSys 2020

Compress, then explain again

Compression and XAI are complementary checks



Average accuracy hides where compression hurts: errors concentrate on rare, atypical and under-represented examples [10, 11].

Checklist after every compression step

- Accuracy and per-class recall, plus the confusion matrix
- Grad-CAM on the same images (on the pruned model: the int8 model has no gradients)
- Size and CPU latency

[10] Hooker et al., arXiv 2019 · [11] Hooker et al., arXiv 2020 · [31] Khalid et al., preprint 2026 (see notes)

Your notebook

Before the workshop: sections 1 to 7. On the workshop day: section 8 with MANOLO

1	2	3	4	5	6	7	8
Setup	Data	Model	Training	Results	XAI	Compression	MANOLO
					Grad-CAM	*int8 · pruning*	*workshop day*

Pick one dataset



Wildfire



Rice blast



Chest X-ray

Bring to the workshop

- Where does your model look?
- Which compression gives the best trade-off?
- What other things can you try?

Tools: pytorch-grad-cam · torch.ao.quantization · torch.nn.utils.prune

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CLOUD-EDGE EFFICIENT & TRUSTWORTHY AI

Thank you!

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Funded by the European Union

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[30] D. Blalock et al. What is the state of neural network pruning? MLSys, 2020.

[31] N. Khalid et al. Understanding the impact of model pruning on long-tail forgetting and explanation reliability in medical imaging. Preprint, 2026.